

Financial Risk Analysis & Loan Portfolio Dashboard for InstaBank

1. Domain Knowledge / Business Problem

➤ Background:

Banks manage thousands of loans across different products such as Business Loans, Car Loans, Home Loans, and Personal Loans. Identifying risky borrowers and monitoring portfolio performance is critical to reducing financial losses and maintaining healthy lending operations.

➤ Business Need:

InstaBank required a centralized dashboard to:

- Monitor loan portfolio performance.
- Identify high-risk customers and default trends.
- Analyze regional and product-level lending risks.
- Track portfolio growth and loan disbursements.
- Support data-driven lending decisions.

➤ Stakeholders:

- Risk Management Team
- Credit Analysts
- Loan Officers
- Regional Managers
- Executive Leadership Team

2. The Challenge (Problem Statement)

The bank faced several analytical challenges:

- No consolidated view of loan portfolio performance.
- Difficult to identify regions with higher default rates.
- Limited visibility into customer credit score risk.
- Manual analysis of default trends was time-consuming.
- Loan officers lacked insights into high-risk borrower segments.

Management required quick visibility into loan performance across multiple dimensions.

3. The Approach (Solution Design)

- **Built an Excel-Based Financial Risk Dashboard with:**

- **Loan Portfolio Analysis**

Analyzed loan distribution across:

- Business Loans
- Car Loans
- Home Loans
- Personal Loans

Total loan portfolio exceeded ₹1 Billion across 1,000 loan records.

- **Default Risk Analysis**

Calculated:

- Total Loans
- Paid Loans
- Default Loans
- Default Percentage

Default Rate = 19.58% of total loans.

- **Credit Risk Segmentation**

Grouped customers by credit score bands:

- 600-649
- 650-699
- 700-749
- 750-799
- 800-850

Used credit score analysis to identify high-risk customer categories and default concentration.

- **Regional Risk Analysis**

Created region-wise default analysis across:

- East
- North
- South
- West

East region recorded the highest number of defaults, helping identify geographical risk concentration.

- **Loan Type Performance**

Compared:

- Loan Type vs Default Status

- Loan Type vs Interest Rate
- Loan Type vs Loan Amount

Identified Personal Loans and Business Loans as key contributors to portfolio risk.

➤ Customer Segmentation

Analyzed portfolio based on:

- Employment Type
- Annual Income
- Credit Score
- Loan Amount
- Loan Status

Provided insights into borrower behavior and repayment patterns.

➤ Trend Analysis

Created yearly loan disbursement analysis from:

- 2018 to 2024

Used trend charts to monitor lending growth and portfolio expansion.

➤ Excel Features Used

- Pivot Tables
- Pivot Charts
- VLOOKUP/XLOOKUP
- IF Functions
- COUNTIFS / SUMIFS
- Conditional Formatting
- Data Validation
- Dashboard KPIs
- Interactive Slicers and Filters

4. The Impact (Results & Benefits)

- Enabled quick identification of high-risk customer segments.
- Improved visibility into default trends across regions.
- Helped management track a portfolio worth over ₹1 Billion efficiently.
- Reduced manual analysis and reporting effort significantly.
- Supported proactive risk mitigation through credit score and region-based monitoring.
- Improved decision-making for loan approval and portfolio management.

5. Reflection (Learning & Future Scope)

➤ Learned

- Financial Risk Analysis concepts.
- Credit Risk Segmentation.

- Loan Portfolio Management.
- Advanced Excel Dashboard Development.
- KPI Design and Data Visualization.
- Banking and Lending Analytics.

➤ Future Improvements

- Build the dashboard in Power BI.
- Automate data refresh using Power Query.
- Add predictive default-risk scoring models.
- Integrate customer transaction history.
- Develop AI-driven risk forecasting dashboards.
- Create automated alerts for high-risk borrowers.